

Guided operating reference

A source-derived sequence for trained, authorised teachers

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Use this guide to prepare with your supplier or authorised trainer and to match each step to the actual installation. It explains the documented sequence, expected states and checks. It does not establish that your machine is commissioned, that these controls match your installed version, or that a person is authorised to operate.

The supplied school equipment guidance reserves operation for trained teachers and says it cannot be delegated. Students may prepare programs before use. Machine simulation still creates real movement. Students must remain outside the controlled work area; entry requires electrical isolation under the supplied guidance.

The main source is the supplied SureControl Software Manual, internal V2.1. Source page references appear with each step. Example screen values are not approved cutting settings.

If the controls, post-processor, toolset, material, consumables or approved settings do not match, stop and obtain the matching local procedure. Never solve a mismatch by copying an example value, trying a different machine configuration or bypassing a safeguard.

How to use each page

Read what you need, follow the documented actions with your trainer, compare the expected state, and perform the stated check. Use the stop point whenever evidence is missing or behaviour is unexpected. Record local differences and the authorised trial result in your school records.

Step 01

Confirm operator and installation readiness

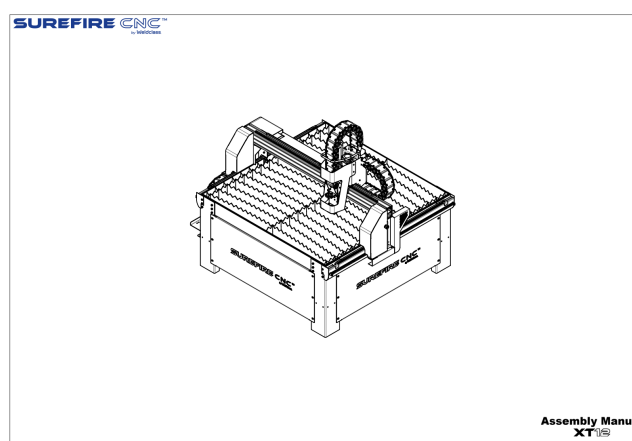
Make sure the guide supports approved supplier training and the actual school installation.

What you need

- Comprehensive supplier training, principal approval and site-specific installation approval.
- Current school operating procedure, emergency-stop method and electrical isolation procedure.
- Confirmed machine, power source, torch, software version, post-processor and material recipe.

Follow these steps

1. Check the school records identify you as the trained teacher authorised to operate this equipment. Completing this course does not provide operator authorisation.
2. Locate the real emergency stop, keyed control, electrical isolation points, extraction controls and access controls. Have their use demonstrated if you cannot identify them confidently.
3. Check that the school procedure matches the installed system and that its required guards, screens, interlocks and extraction are serviceable.
4. While an operating prerequisite is unresolved, continue only the offline file-preparation stages. Do not connect to, energise or move the machine.



Supplier XT12 overview from the assembly manual, p1. This is a machine drawing, not a photograph of the school installation or its safety enclosure.

What you should see

An authorised operator and approved installation, with the current school procedure available beside the guide.

How to check

The supplied Department policy permits students to help with programming before use. Operating this equipment is teacher-only and cannot be delegated.

Stop and seek help: Stop before machine operation if training, approvals, controls or equipment identity are uncertain. Resolve this with the school and supplier.

Documentation reference: CNC Plasma.pdf pp1-2; SureControl pp55-62. Match the current school installation before use.

[Open the supporting source](#)

Step 02

Accept the correct drawing revision

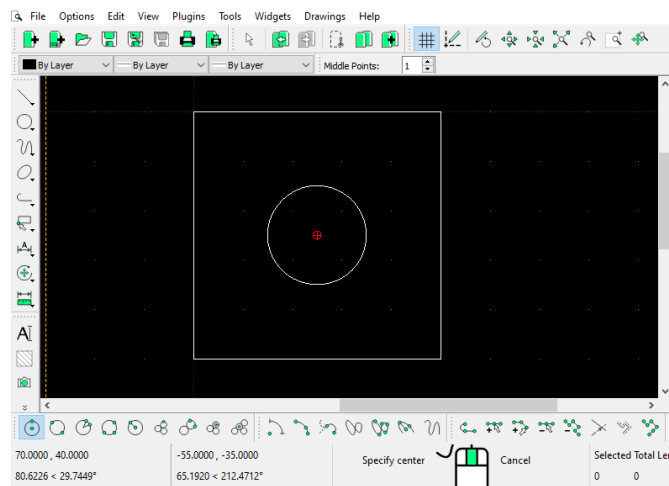
Begin with a checked design and clear physical dimensions.

What you need

- The editable CAD master and its separately exported DXF.
- A drawing record giving intended dimensions, units and revision.

Follow these steps

1. Identify the approved DXF filename and revision. Preserve the editable original.
2. Open the DXF in the chosen offline design/CAM pathway and check the outline and every hole.
3. Check imported units and dimensions. Inspect gaps, repeated lines, unsupported curves and small details that will not remain attached.
4. Resolve drawing problems and export a new named revision before production preparation.



The manual uses a simple square and hole to illustrate the file journey. Source: SureControl manual V2.1, page 14. Supplier image; current school version unverified and shown values are examples.

What you should see

One identified design whose imported size and contours agree with its drawing record.

How to check

Fitting on screen does not prove size. Compare the overall dimensions and important hole dimensions.

Stop and seek help: Return to CAD if geometry, size, units or revision disagree. Do not hide a drawing problem with cutting settings.

Documentation reference: SureControl pp14-15,35. Match the current school installation before use.

Open the supporting source

Step 03

Import into SheetCam and inspect the operation

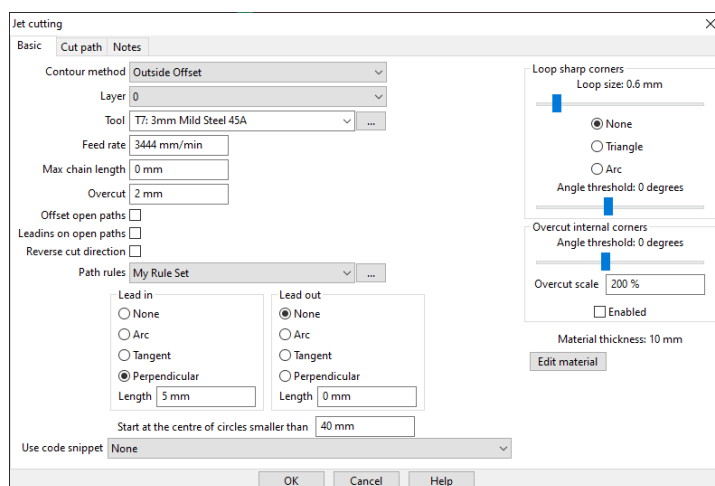
Prepare the cutting path using the verified school CAM setup.

What you need

- Verified SheetCam configuration, post-processor and toolset for the fitted source.
- Approved material, thickness, consumables and cutting chart.

Follow these steps

1. Choose File > Import drawing in SheetCam and select the checked DXF. The source Drawing options screenshot uses Metric and a bottom-left drawing position; confirm these match this job.
2. Choose the jet-cutting operation and relevant layer. Select the verified tool for the actual material, thickness and consumables.
3. Set the approved contour method and lead-in/out strategy. Inspect hole paths and the outside path separately, including which side retains the finished part.
4. Display toolpaths, path directions, start points, rapid moves and path ends. Check the complete job and all travel between cuts.
5. Run SheetCam's on-screen simulation. This offline preview is different from SureControl's real-machine Simul mode.



Locate the contour method, layer, tool and lead-in controls. Source: SureControl manual V2.1, page 16. Supplier image; current school version unverified and shown values are examples.

What you should see

A complete CAM job with the intended paths and no unresolved import or processing problems.

How to check

The manual's sample layer 0, My Rule Set, tool and lead-in values are examples. Use the validated school recipe and inspect the resulting paths.

Stop and seek help: Stop if a contour is missing, a lead-in damages retained material, travel leaves the sheet or the tool/post-processor is uncertain.

Documentation reference: SureControl pp15-16,41-44,48-50. Match the current school installation before use.

Open the supporting source

Step 04

Create and transfer the TAP file

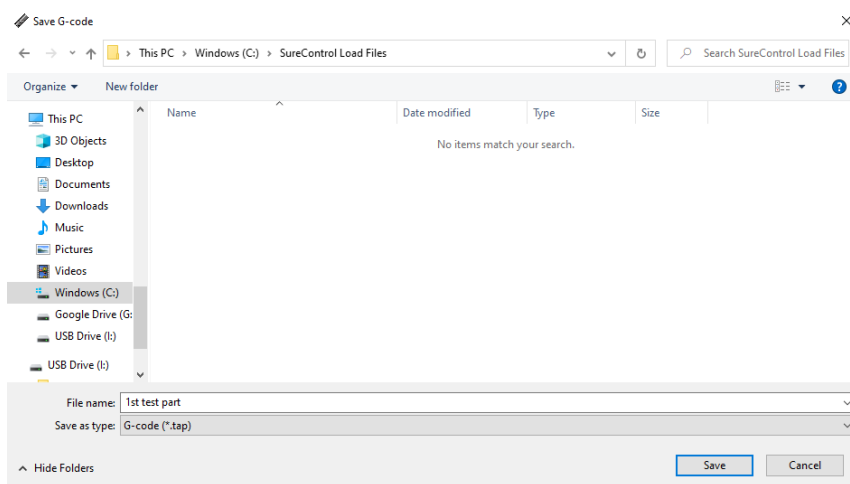
Keep the drawing, editable CAM job and machine program distinct.

What you need

- Checked CAM job and the verified SureControl post-processor.
- Agreed transfer method and confirmed machine-computer folder.

Follow these steps

1. Save the editable CAM job so it can be revised later.
2. Run the post-processor using the toolbar command or File > Run post processor. Review its completion messages.
3. Save the G-code output with its .tap extension and a clear revision. The supplier example uses C:\SureControl Load Files; confirm the actual destination.
4. If preparing on another computer, transfer that exact TAP revision by the school's approved method.
5. Check the destination filename, extension and revision before loading it. Preserve the DXF and CAM master separately.



Confirm the real destination folder and file revision. Source: SureControl manual V2.1, page 17. Supplier image; current school version unverified and shown values are examples.

What you should see

A successfully processed TAP file traceable to the checked design, in the confirmed machine-computer folder.

How to check

DXF is the exchange drawing; TAP is the machine-specific program. Renaming a file does not convert it.

Stop and seek help: Stop for post-processing errors, an unconfirmed post-processor or an ambiguous destination file.

Documentation reference: SureControl pp9,11,17,46. Match the current school installation before use.

Open the supporting source

Step 05

Prepare the controlled work area

Put approved separation, fume and fire controls in place before movement or cutting.

What you need

- Stage 1 completed and the school pre-use procedure.
- Students outside the controlled work area and an approved viewing arrangement.
- An approved operator position physically separated from the machine by the securely screened operating area and welding curtains during automatic cutting.

Follow these steps

1. Apply the school pre-use checks to the purpose-built dry operating environment, slag collection drawer, guards covering exposed drive belts and pinch points, screens and welding curtains, access controls and mechanical fume extraction. Confirm the required items are in place and serviceable.
2. Confirm the operator is also physically separated from the machine by the securely screened operating area and welding curtains during automatic cutting. Teacher-only operation does not permit standing inside the enclosure while the machine cuts.
3. Remove flammable material and obstructions under the school procedure. Plan safe sheet handling and the production queue.
4. Confirm the material is approved and has no prohibited coating or unidentified contamination.
5. Keep students isolated from the operating area. They may enter the shielded area only after the required electrical isolation.

What you should see

A prepared area with the specified extraction, guarding, separation and controlled access.

How to check

Check both student separation and the operator's screened position. Manufacturer assembly drawings do not demonstrate that the school's enclosure, extraction or installation approval is adequate.

Stop and seek help: Do not operate with a missing, damaged or uncertain slag collection drawer, guard, access control, screen, extraction or isolation control, or in an unsuitable wet environment. Report the fault.

Documentation reference: CNC Plasma.pdf pp1-2; SureControl pp55-62. Match the current school installation before use.

[Open the supporting source](#)

Step 06

Match the physical setup and recipe

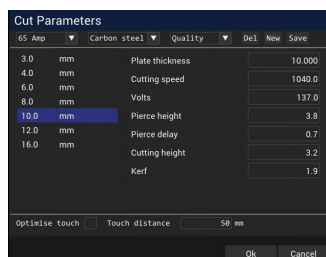
Match the sheet, torch and physical settings to the software recipe.

What you need

- Authorised operator; prepared controlled area.
- Approved material recipe and the correct power-source/torch documentation.

Follow these steps

1. Load and support the selected sheet using the school procedure.
2. Before power-on, identify and check the work-return lead and clamp. Use the supplier/site-approved connection point and verify the connection and contact against the matched power-source instructions. Do not choose an unverified return path.
3. Verify the controller-to-power-source interface/CPC connection against the approved installation record. Leave wiring or configuration faults to authorised support.
4. Check the fitted consumables, approved connections and machine condition against supplier pre-use requirements. Do not service or adjust hardware while energised.
5. Check air supply and any fitted water tray against the approved requirements. The software manual does not establish this school's pressure or fill level.
6. Set the required current on the physical plasma power source using its approved procedure. Match it to the software recipe and fitted consumables.
7. Apply the school's approved power-on sequence for the source, controller and extraction, including any fitted torch-enable switch.



The software Amp selection must match the separately set physical power-source current. Source: SureControl manual V2.1, page 27. Supplier image; current school version unverified and shown values are examples.

What you should see

The sheet, torch parts, physical current, air supply and software recipe agree.

How to check

Selecting Amp in SureControl selects software parameters. It does not change the physical power-source amperage. Confirm the work-return lead and clamp have been checked, not merely assumed to be included in general connections.

Stop and seek help: Stop if the material, parts, current, supply or connections cannot be matched to the approved recipe. Do not substitute screenshot values.

Documentation reference: SureControl pp13,27-28,53; SureControl V2.1 pp53, 61-62: work return and clamp. Match the current school installation before use.

Open the supporting source

Step 07

Connect to the controller

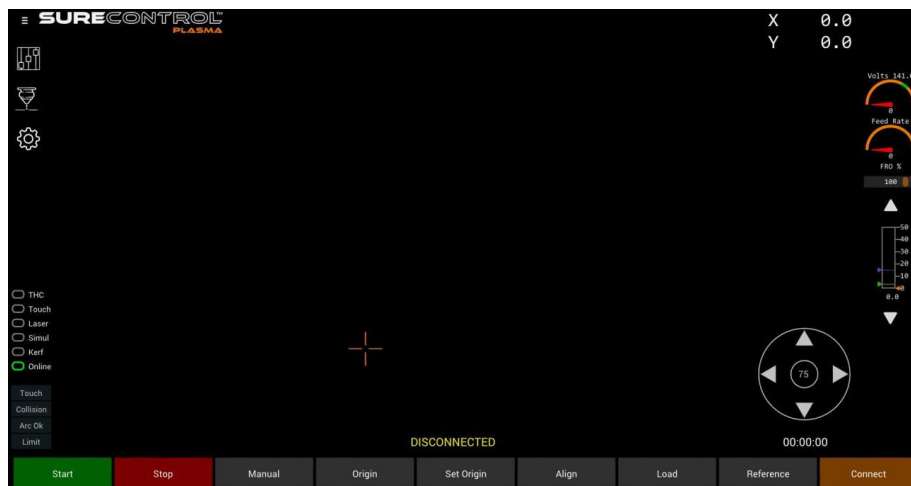
Establish communication and deliberately enable the prepared system.

What you need

- All operating prerequisites completed.
- Confirmed controller connection and unobstructed motion area.

Follow these steps

1. Open SureControl and read the state near the bottom of the screen. The manual describes the initial state as Disconnected.
2. Select Connect. The source says the first connected state is Offline, which inhibits motion and torch activation in software.
3. Check the state and messages against the approved setup. Only when ready, select Online as taught during supplier training.
4. Expect Not Referenced if the axes still require referencing. Continue using the validated school routine.



SureControl main screen with state text and Connect button Source: SureControl manual V2.1, page 25. Supplier image; current school version unverified and shown values are examples.

What you should see

The state advances through the documented connection sequence without unexplained messages or movement.

How to check

Disconnected, Offline and Not Referenced mean different things. Software Offline does not establish electrical isolation.

Stop and seek help: Stop for a failed connection, unexpected state or unintended motion. Use the approved procedure; do not experiment with network/controller configuration.

Documentation reference: SureControl pp6,30-33. Match the current school installation before use.

[Open the supporting source](#)

Step 08

Reference the machine

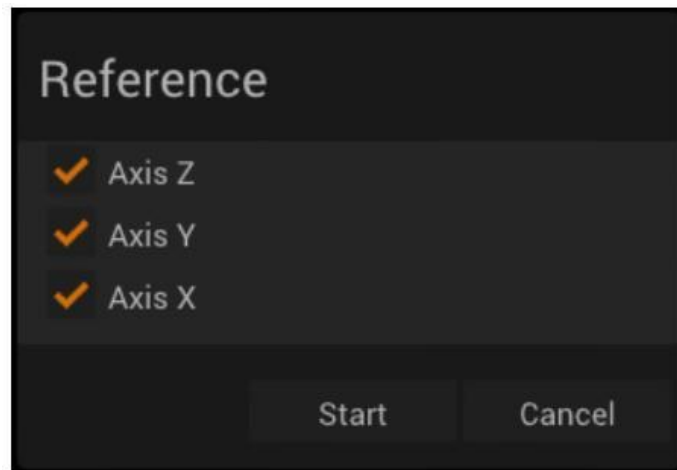
Give the controller a known machine position.

What you need

- Validated referencing method and clear motion area.
- Authorised teacher at the controls with the emergency-stop method understood.

Follow these steps

1. Select Reference to open the axis-selection dialog.
2. Confirm the selected axes match the supplier-approved routine. The manual's initial setup selects all axes; its overview describes Z, then Y, then X.
3. Select Start only when ready for real movement. Observe the referencing motion from the approved position.
4. After normal completion, close the dialog as directed and check the reported position/state.



Start in this dialog causes real movement. Source: SureControl manual V2.1, page 30. Supplier image; current school version unverified and shown values are examples.

What you should see

The required axes reference normally and the controller recognises their machine positions.

How to check

Referencing does not place the design on the sheet or set the part origin. The software's minimum requirement is not a replacement for the school routine.

Stop and seek help: If an axis travels the wrong way, is obstructed or fails to reference, use Cancel or the emergency stop as appropriate. Seek supplier help; do not change direction polarity, sensor inputs or limits.

Documentation reference: SureControl pp6,30,33. Match the current school installation before use.

[Open the supporting source](#)

Step 09

Load and identify the production program

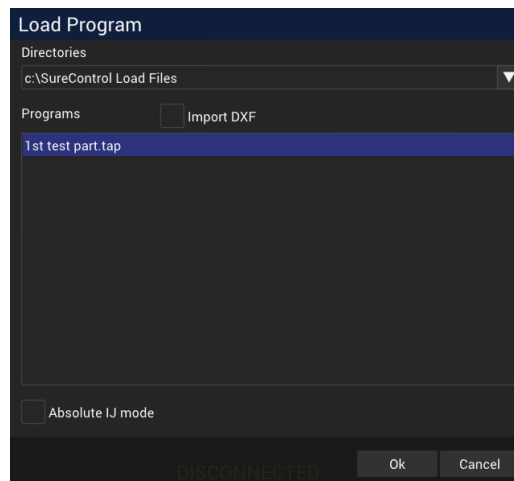
Confirm that the displayed program is the checked job.

What you need

- Checked TAP revision in the confirmed folder.
- Machine referenced according to the approved routine.

Follow these steps

1. Select Load to open Load Program.
2. Choose the confirmed directory and correct .tap filename. For this route, do not switch to Import DXF accidentally.
3. Select OK. Compare the displayed profile, holes and orientation with the design/CAM record.
4. Read the state. The manual describes Set Origin after a program is loaded and its origin has not yet been assigned.



Load Program dialog listing a TAP file Source: SureControl manual V2.1, page 18. Supplier image; current school version unverified and shown values are examples.

What you should see

The correct program appears with its intended profile and holes.

How to check

Compare your own checked revision with the screen. A plausible-looking preview does not prove file identity.

Stop and seek help: Stop for a wrong file, missing shape or unexplained difference. Resolve it in file preparation before continuing.

Documentation reference: SureControl pp18,30,33. Match the current school installation before use.

Open the supporting source

Step 10

Check material-height sensing

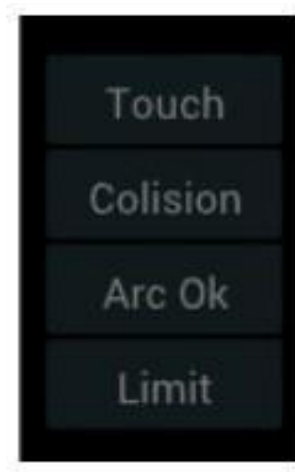
Confirm that the fitted sensing system recognises the material.

What you need

- Supplier training in the actual sensing method.
- Approved pre-cut check, positioned sheet and clear motion area.

Follow these steps

1. Use the validated sensor check for this installation. The supplied manual describes gently lowering the torch with Page Down until sensing contact is made.
2. During that approved check, observe the Touch status indicator. The source expects it to turn green when contact is detected.
3. Distinguish the status indicator from the Touch option that enables sensing. Check the option matches the approved plasma setup.
4. Return to the appropriate safe position using the supplier-taught method.



Status indicators are separate from the option switches. Source: SureControl manual V2.1, page 31. Supplier image; current school version unverified and shown values are examples.

What you should see

The sensor response agrees with contact and no unexpected collision indication occurs.

How to check

The Touch indicator is not an instruction to touch the torch by hand. A green option does not prove the sensor has operated.

Stop and seek help: Stop for failed indication, unexpected pressing or an unclear sensing method. Isolate as required and seek support; do not bypass it by changing input numbers, offsets or limits.

Documentation reference: SureControl pp19,31,40. Match the current school installation before use.

[Open the supporting source](#)

Step 11

Complete the approved torch check

Check torch readiness by the method the trained teacher is authorised to use.

What you need

- Approved torch-test method, controlled area and required protection.
- Correct source, air and interface conditions confirmed.

Follow these steps

1. Confirm whether the current school procedure requires the manual's test-fire check here. If it differs, follow the approved procedure.
2. Locate the torch-shaped Torch On/Off icon. It fires the actual torch; it is not a preview.
3. Under the approved conditions, perform the supplier-taught test. The manual describes selecting the icon to fire and selecting it again to turn the test fire off.
4. Verify the test has ended using the approved method. Read any power-source fault indication before proceeding.



This control can fire the real torch. Source: SureControl manual V2.1, page 19. Supplier image; current school version unverified and shown values are examples.

What you should see

The torch behaves as expected during the authorised check and is no longer firing afterwards.

How to check

A successful test does not replace the checks of program, origin, material, extraction or safe access.

Stop and seek help: Stop if behaviour is unexpected, the torch fails to turn off or a fault appears. Apply the school emergency/fault procedure; do not repeat firing to see whether it clears.

Documentation reference: SureControl pp19,29,53. Match the current school installation before use.

[Open the supporting source](#)

Step 12

Position the part and set its origin

Place the checked design on the sheet using its intended part zero.

What you need

- Loaded program and approved origin strategy.
- Teacher-controlled jogging method and known sheet position.

Follow these steps

1. Identify where the design's origin belongs on the sheet. The supplier example uses the front/bottom-left position; confirm the agreed orientation for this job.
2. Jog to the intended origin using the approved method and observing clearance.
3. Select Set Origin to assign the current position as the part origin.
4. Check the displayed coordinates, orange crosshairs and relationship between the part, lead-ins and available sheet.



Set Origin assigns a position; Origin commands movement to it. Source: SureControl manual V2.1, page 19. Supplier image; current school version unverified and shown values are examples.

What you should see

The part origin and displayed layout agree with the approved job.

How to check

Set Origin assigns the current location as part zero. Origin is different: it commands the machine to move to that origin.

Stop and seek help: Stop if the part zero is unclear or the complete job cannot fit. Recheck the layout before cutting.

Documentation reference: SureControl pp19,30,33. Match the current school installation before use.

[Open the supporting source](#)

Step 13

Run the approved machine dry run

Check actual travel with the torch disabled by simulation mode.

What you need

- Checked program and origin; all motion/access controls in force.
- Authorised teacher ready to stop unexpected motion.

Follow these steps

1. Confirm the approved dry-run conditions. This stage uses the real machine.
2. Select Simul, then Start as described in the manual.
3. Observe real travel from the approved position. Check the sequence, placement and travel clearance against the job.
4. At the end, verify completion. Keep cutting disabled while resolving anything unexpected.



Simul is a real-machine dry run. Source: SureControl manual V2.1, page 19. Supplier image; current school version unverified and shown values are examples.

What you should see

The machine follows the programmed sequence without firing the torch.

How to check

SureControl Simul moves the real CNC. SheetCam's offline on-screen preview does not. Students must not operate the connected machine for this check.

Stop and seek help: Stop if the torch fires, motion is unexpected, an obstruction exists or travel does not fit. Resolve the cause before enabling cutting.

Documentation reference: SureControl pp19,31; contrast p42. Match the current school installation before use.

[Open the supporting source](#)

Step 14

Confirm cutting mode and produce the job

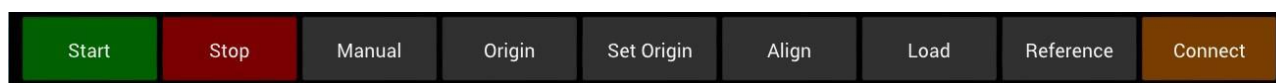
Make a deliberate final check before a real cut.

What you need

- Completed approved checks and validated recipe/mode settings.
- Secure controlled area and operating extraction, with the teacher at the approved controls physically separated from the machine by the securely screened operating area and welding curtains.

Follow these steps

1. Recheck material/thickness recipe, physical current and fitted consumables against the job record.
2. Check every option against the validated school route. The supplied manual requires Touch enabled for plasma and identifies Laser as unused in its version.
3. Confirm Simul is off when intentionally cutting. Confirm Online, THC and Kerf match the approved recipe and CAM route; do not copy screenshot flags blindly.
4. Recheck program identity, origin and controlled access. Confirm the teacher is at the approved operator position, physically separated from the machine by the securely screened operating area and welding curtains. Select Start only after these checks pass.
5. Observe the automatic cut from that screened operator position and remain physically separated from the machine throughout cutting. Do not stand inside the enclosure. Remain ready to use the school stop/emergency procedure.



Software Stop does not by itself establish electrical isolation. Source: SureControl manual V2.1, page 30. Supplier image; current school version unverified and shown values are examples.

What you should see

The machine executes the intended cut under approved controls and settings.

How to check

The manual's first-cut flags are one example, not an independently validated school recipe. Confirm where compensation is applied and how THC is used for this job.

Stop and seek help: Stop for unexpected motion, arc loss, faults, extraction failure or unsafe access. After interruption, do not select a resume point casually; recovery requires the trained operator's assessment.

Documentation reference: SureControl pp20,27,30-32; CNC Plasma.pdf p1. Match the current school installation before use.

Open the supporting source

Step 15

Finish, isolate, retrieve and record

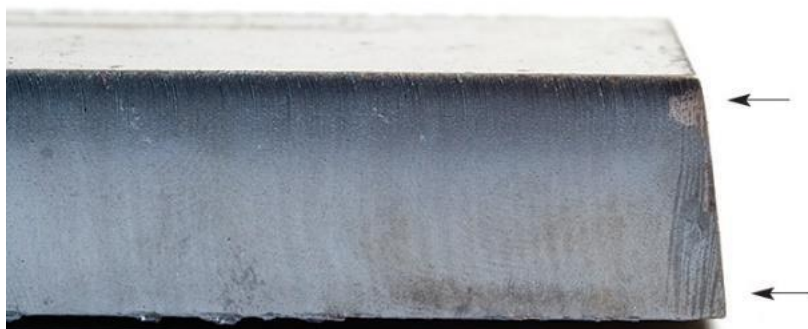
Leave the equipment safe and preserve evidence of the result.

What you need

- Approved end-of-cut, cooling, isolation and handling procedure.
- Suitable handling equipment/protection and an inspection record.

Follow these steps

1. Confirm the program has finished and read the machine state. Idle is a software state, not proof of isolation.
2. Complete the approved shutdown/isolation sequence. The source requires power off when unattended and disconnection before inspection, maintenance or cleaning.
3. Before any student enters the shielded area, confirm electrical isolation using the school procedure.
4. Retrieve material only when approved cooling and handling conditions are satisfied. Parts and stock can remain hot or sharp.
5. Measure the finished part and inspect its edge. Record file revision, recipe identifier, observations and faults for the trained operator/supplier.



Positive bevel reference photograph, SureFire CNC cut-quality guidance p1. Supplier example, not the school's cutting result.

What you should see

The machine is in its approved safe state, material is handled safely and the result is traceable.

How to check

Program completion, Stop, Offline or an extinguished arc does not replace electrical isolation.

Stop and seek help: Do not reach in or allow entry until isolation and handling conditions are satisfied. Do not clean or service energised equipment.

Documentation reference: CNC Plasma.pdf p2; SureControl pp32,51-56,61-63. Match the current school installation before use.

[Open the supporting source](#)